TWO TENSOR FUNCTORS WITH ISOMORPHIC PULLBACKS GIVE 2-ISOMORPHIC MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4791-4799 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4801-4845, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 32 of queue_stacks-stacks-more-morphisms.json (pb=1705, stmt 4791-4799, proof 4801-4845).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: 'Some details omitted' for the final descent chase - bookkeeping on the displayed identification (artin #32 precedent).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-fully-faithful' also exists in stacks-descent — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4575–4583. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04YC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4845–4845. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack with affine diagonal. Let B be a ring. Let $f_i : \mathrm{Spec}(B) \to \mathcal{X}$, $i = 1, 2$ be two morphisms. Let $t : f_1^* \to f_2^*$ be an isomorphism of the tensor functors $f_i^* : QCoh(\mathcal{O}_\mathcal{X}) \to Mod_B$. Then there is a 2-arrow $f_1 \to f_2$ inducing t.*

Proof. Choose an affine scheme $U = \mathrm{Spec}(A)$ and a surjective smooth morphism $g : U \to \mathcal{X}$, see Properties of Stacks, Lemma 04YC. Since the diagonal of $\mathcal{X}$ is affine, we see that $U_i = \mathrm{Spec}(B) \times_{f_i, \mathcal{X}, g} U$ is affine. Say $U_i = \mathrm{Spec}(C_i)$. Then C_i is the B-algebra endowed with ring map $A \to C_i$ constructed in Lemma lemma-affine using the functor $F = f_i^*$. Therefore t induces an isomorphism $C_1 \to C_2$ of B-algebras, compatible with the ring maps $A \to C_1$ and $A \to C_2$. In other words, we have a commutative diagrams

$$\begin{array}{ccc} U_i & \longrightarrow & U \\ \downarrow & & \downarrow{\scriptstyle g} \\ \mathrm{Spec}(B) & \xrightarrow{f_i} & \mathcal{X} \end{array} \qquad \begin{array}{ccccc} & & U_2 & & \\ & \swarrow & \downarrow{\scriptstyle \cong} & \searrow & \\ \mathrm{Spec}(B) & \longleftarrow & U_1 & \longrightarrow & U \end{array}$$

This already shows that the objects f_1 and f_2 of $\mathcal{X}$ over $\mathrm{Spec}(B)$ become isomorphic after the smooth covering $\{U_1 \to \mathrm{Spec}(B)\}$. To show that this descends to an isomorphism of f_1 and f_2 over $\mathrm{Spec}(B)$, we have to show that our isomorphism (which comes from the commutative diagrams above) is compatible with the descent data over $U_1 \times_{\mathrm{Spec}(B)} U_1$. For this we observe that $U \times_\mathcal{X} U$ is affine too, that we have the morphism $g' : U \times_\mathcal{X} U \to \mathcal{X}$, and that

$$U_i \times_{\mathrm{Spec}(B)} U_i = \mathrm{Spec}(B) \times_{f_i, \mathcal{X}, g'} (U \times_\mathcal{X} U)$$

It follows that the isomorphism $C_1 \otimes_B C_1 \to C_2 \otimes_B C_2$ coming from the isomorphism $C_1 \to C_2$ is compatible with the morphisms $U_i \times_{\mathrm{Spec}(B)} U_i \to U \times_\mathcal{X} U$. Some details omitted. □